Python Programming: Experiment 01

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Question 1: Python Installation and Difference Between Modes

Steps to Install Python:

1. Go to https://www.python.org/downloads/
2. Click on Download Python.
3. Open the downloaded file.
4. Tick the option "Add Python to PATH".
5. Click Install Now.
6. Wait for installation to complete.
7. Click Close.

Difference Between Interactive Mode and Script Mode

Interactive Mode:

- Runs one line at a time.
- Shows output immediately.
- Used for testing small code.

Script Mode:

- Write full program in a file.
- Save the file.
- Run the program using Run option.
- Used for full programs.

Question 2: Create a variable to store age and print its type

```
age = 18
print("Age:", age)
print("Type of age:", type(age))
```

```
Age: 18
Type of age: <class 'int'>
```

Question 3: Declare string variable x and print it

```
x = "Hello"
print("Value of x:", x)
```

```
Value of x: Hello
```

Question 4: Different Data Types

```
a = 10
b = 3.5
c = "Python"
d = True

print(a)
print(b)
print(c)
print(d)
```

```
10
3.5
Python
True
```

Question 5: Arithmetic Operations

```
x = 9
y = 7

print("Addition:", x + y)
print("Multiplication:", x * y)
print("Division:", x / y)
print("Subtraction:", x - y)
```

```
Addition: 16
Multiplication: 63
Division: 1.2857142857142858
Subtraction: 2
```

Question 6: Hypotenuse using Pythagoras Theorem

```
a = 3
b = 4

c = (a*a + b*b) ** 0.5
print("Hypotenuse:", c)
```

```
Hypotenuse: 5.0
```

Question 7: Simple Interest

```
P = 1000
R = 5
T = 2

SI = (P * R * T) / 100
print("Simple Interest:", SI)
```

```
Simple Interest: 100.0
```

Question 8: Area of Triangle

```
a = 3
b = 4
c = 5

s = (a + b + c) / 2
area = (s*(s-a)*(s-b)*(s-c)) ** 0.5

print("Area:", area)
```

```
Area: 6.0
```

Question 9: Convert Seconds into Hours, Minutes and Seconds

```
seconds = 3665

hours = seconds // 3600
minutes = (seconds % 3600) // 60
remaining_seconds = seconds % 60

print("Hours:", hours)
```

```
print("Minutes:", minutes)
print("Seconds:", remaining_seconds)
```

```
Hours: 1
Minutes: 1
Seconds: 5
```

Question 10: Swap Two Numbers Without Extra Variable

```
a = 5
b = 10

a, b = b, a

print("a =", a)
print("b =", b)
```

```
a = 10
b = 5
```

Question 11: Sum of First n Natural Numbers

```
n = 5
sum_n = n * (n + 1) // 2
print("Sum:", sum_n)
```

```
Sum: 15
```

Question 12: Bitwise Operators

```
a = 5
b = 3

print("AND:", a & b)
print("OR:", a | b)
print("XOR:", a ^ b)
```

```
AND: 1
OR: 7
XOR: 6
```

Question 13: Left Shift and Right Shift

```
num = 8

print("Left Shift:", num << 1)
print("Right Shift:", num >> 1)
```

```
Left Shift: 16
Right Shift: 4
```

Question 14: Membership Operator

```
sequence = (10,20,56,78,89)
num = 56

print("Is number in sequence?", num in sequence)
```

```
Is number in sequence? True
```